

Technical Documentation

Diesel Engine
12 V 956 TB31
Generating Set

Description and
Operating Instructions
M010410/00E



Printed in Germany

© 2006 Copyright MTU Friedrichshafen GmbH

Diese Veröffentlichung einschließlich aller ihrer Teile ist urheberrechtlich geschützt. Jede Verwertung oder Nutzung bedarf der vorherigen schriftlichen Zustimmung der MTU Friedrichshafen GmbH. Das gilt insbesondere für Vervielfältigung, Verbreitung, Bearbeitung, Übersetzung, Mikroverfilmungen und die Einspeicherung und / oder Verarbeitung in elektronischen Systemen, einschließlich Datenbanken und Online-Diensten. Das Handbuch ist zur Vermeidung von Störungen oder Schäden beim Betrieb zu beachten und daher vom Betreiber dem jeweiligen Wartungs- und Bedienungspersonal zur Verfügung zu stellen. Änderungen bleiben vorbehalten.

Printed in Germany

© 2006 Copyright MTU Friedrichshafen GmbH

This Publication is protected by copyright and may not be used in any way whether in whole or in part without the prior written permission of MTU Friedrichshafen GmbH. This restriction also applies to copyright, distribution, translation, microfilming and storage or processing on electronic systems including data bases and online services.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation. Subject to alterations and amendments.

Imprimé en Allemagne

© 2006 Copyright MTU Friedrichshafen GmbH

Tout droit réservé pour cet ouvrage dans son intégralité. Toute utilisation ou exploitation requiert au préalable l'accord écrit de MTU Friedrichshafen GmbH. Ceci s'applique notamment à la reproduction, la diffusion, la modification, la traduction, l'archivage sur microfiches, la mémorisation et / ou le traitement sur des systèmes électroniques, y compris les bases de données et les services en ligne.

Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite.

Modifications réservées.

Impreso en Alemania

© 2006 Copyright MTU Friedrichshafen GmbH

Esta publicación se encuentra protegida, en toda su extensión, por los derechos de autor. Cualquier utilización de la misma, así como su reproducción, difusión, transformación, traducción, microfilmación, grabación y/o procesamiento en sistemas electrónicos, entre los que se incluyen bancos de datos y servicios en línea, precisa de la autorización previa de MTU Friedrichshafen GmbH.

El manual debe tenerse presente para evitar fallos o daños durante el servicio, y, por dicho motivo, el usuario debe ponerlo a disposición del personal de mantenimiento y de servicio.

Nos reservamos el derecho de introducir modificaciones.

Stampato in Germania

© 2006 Copyright MTU Friedrichshafen GmbH

Questa pubblicazione è protetta dal diritto d'autore in tutte le sue parti. Ciascun impiego o utilizzo, con particolare riguardo alla riproduzione, alla diffusione, alla modifica, alla traduzione, all'archiviazione in microfilm e alla memorizzazione o all'elaborazione in sistemi elettronici, comprese banche dati e servizi on line, deve essere espressamente autorizzato per iscritto dalla MTU Friedrichshafen GmbH.

Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

Con riserva di modifiche.

Impresso na Alemanha

© 2006 Copyright MTU Friedrichshafen GmbH

A presente publicação, inclusive todas as suas partes, está protegida pelo direito autoral. Qualquer aproveitamento ou uso exige a autorização prévia e por escrito da MTU Friedrichshafen GmbH. Isto diz respeito em particular à reprodução, divulgação, tratamento, tradução, microfilmagem, e a memorização e/ou processamento em sistemas eletrônicos, inclusive bancos de dados e serviços on-line.

Para evitar falhas ou danos durante a operação, os dizeres do manual devem ser respeitados. Quem explora o equipamento economicamente consequentemente deve colocá-lo à disposição do respetivo pessoal da conservação, e à disposição dos operadores.

Salvo alterações.

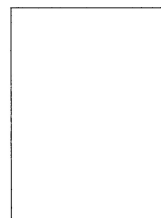
Amendment Service!

Any changes in contents will be sent to you in the form of an amendment,
provided you complete the reverse of this receipt card and return it to us.

M010410/00E

Description and
Operating Instructions

12V 956 TB31



Postcard

MTU Friedrichshafen GmbH
Abteilung MSD
88040 Friedrichshafen
GERMANY

Please mark all filed amendments here.



No.	Date	Name
1.	03.98	
2.	02.01	
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
11.		
12.		
13.		
14.		
15.		
16.		

Please use block capitals!



Receipt

Amendment No.

Tick one box only!

- ☐ Update as (PDF) file by E-mail
☐ Update as (PDF) file as CD-ROM
☐ Update in hardcopy form

Publication number: **M010410/00E**

Name

Manufacturer

Department E-Mail

Fax Telephone

Street

Postal box number

(Postal code) City.....

Country

1.	03.98
2.	02.01
3.	
4.	
5.	
6.	
7.	
8.	
9.	
10.	
11.	
12.	
13.	
14.	
15.	
16.	

LEGAL UNITS OF MEASUREMENT WITH MTU-F

Length, Area, Volume	Unit	SI-Symbol	further permissible symbols (W.E.) hitherto customary symbols etc.
	Length	m	<u>Basic unit</u> W.E.: μ m, mm, km the hitherto "Micron" and μ for 0.001 mm not permitted anymore
	Area [A]	m ²	W.E.: mm ² , a (Ar) = 100 m ² , ha Abbreviations with q, for ex. qm only per- mitted till 12. 1974
	Volume	m ³	W.E.: l (litre), mm ³ , cm ³ , dm ³ Abbreviations cbm, ccm, cmm only permitted till 12. 1974
Angle			
Mass	Plane angle	rad * <i>Radian</i>	1 rad = 1m (arc) : 1 m (radius) W.E.: degree (°) minute (') second (")
	Mass	kg <i>Kilogramme</i>	<u>Basic unit</u> W.E.: mg, g, t hitherto kgs ² /m and kps ² /m Alteration of numerical value <u>Weight</u> statements further as hitherto used in g, kg, t when understood as mass
	Density	kg/m ³	W.E.: g/cm ³ , kg/l hitherto: kgs ² /m ⁴ and kps ² /m ⁴ Alteration of numerical values
	Mass moment 2nd grade <i>Moment of inertia</i>	kg m ²	hitherto: kgms ² and kpms ² Alteration of numerical values
Time			
	Time <i>Periodic time</i>	s <i>Second</i>	<u>Basic unit</u> W.E.: minute (min.), hour (h), day (d) with min., h, d no decimal prefixes do not use as span: week, month, year do not write: sec., sek., Sek, Min., Min, St, Std.
	Frequency	Hz <i>Hertz</i>	1 Hz = 1/s W.E.: kHz, MHz
	Velocity	m/s	W.E.: km/h
	Acceleration [a]	m/s ²	
	Angular velocity	rad/s	W.E.: °/s
	Angular acceleration	rad/s ²	
	Speed	-	rpm, 1/min, 2 π rad/min, rev/min for example: n = 800 rpm, n = 500/min.
	Impulse (momentum)	Ns	hitherto kgs, kps Alteration of numerical values
	Torsional momentum <i>Twist</i>	Nsm	hitherto kgsm, kpsm Alteration of numerical values
	Specific fuel consumption	kg/J*	W.E.: g/kWh hitherto g/PSh Alteration of numerical values

Force, Energy, Power	Unit	SI-Symbol	further permissible symbols (W.E.) hitherto customary symbols etc.
Force [F]	N <i>Newton</i>		1 N = 1 kgm/s ² W.E.: mN, kN, MN hitherto: g, kg, t and p, kp, Mp Alteration of numerical values 1 kp = 9.80665 N ≈ 10 N, 1 Mp ≈ 10 kN etc. for <u>forces of weight</u> caused by masses also N, mN, kN, MN are valid
Pressure <i>Especially of fluids</i>	Pa * <i>Pascal</i>		1 Pa = 1 N/m ² W.E.: <u>bar</u> (Bar), mbar 1 bar = 10 ⁵ Pa = 10 N/cm ² = 0.1 N/mm ² = 1.01972 at hitherto: kg/cm ² , kp/cm ² , at, atm, ata, atü, mmWS, Torr, mm Hg (QS) Alteration of numerical values 1 kp/cm ² = 1 at = 0.980665 bar 1 mmWS = 0.0981 mbar 1 mmHg (QS) = 1 Torr = 1.333 mbar if overpressure, differential pressure, underpressure or absolute pressure <u>differentiate by using the respective terms</u>
Mech. stress, strength, elasticity-, thrust-, <i>(slide) module</i>	N/m ² *		W.E.: N/mm ² hitherto: kg/mm ² , kp/mm ² Alteration of numerical values 1 kp/mm ² = 9.81 N/mm ² ≈ 10 N/mm ² (1 N/mm ² = 0.102 kp/mm ² ≈ 0.1 kp/mm ²) Example: 40 kp/mm ² ≈ 392 N/mm ²
Energy Work Force	J <i>Joule</i>		1 J = 1 Nm = 1 Ws = 1 kgm ² /s ² hitherto: kJm, kpm Alteration of numerical values 1 kpm ≈ 9.81 J
Quantity of heat			W.E.: kJ hitherto: cal, kcal Alteration of numerical values 1 kcal ≈ 4.187 kJ ≈ 1.163 Wh
Momentum of force, torque <i>bending moment</i>	Nm J*		hitherto: kJm, kpm Alteration of numerical values 1 kpm ≈ 9.81 Nm
Notch impact strength	J/m ² *		W.E.: J/mm ² , J/cm ² hitherto: kgm/cm ² , kpm/cm ² Alteration of numerical values
Power [P] Energy flow Heat flow	W* <i>Watt</i>		1 W = 1 J/s = 1 Nm/s W.E.: kW, kJ/h hitherto: PS, kgm/s, kpm/s Alteration of numerical values 1 PS = 75 kpm/s = 75.9.81 Nm/s = 0.736 kW
Viscometrical values			
Dynamic viscosity	Pas		1 Pas = 1 Ns/m ² = 1 kg/ms W.E.: mPas hitherto: P (<i>Poise</i>) 1 P = 0.0102 kps/m ² cP (<i>Centipoise</i>) 1 cP = 1mPas
Kinematic viscosity	m ² /s*		W.E.: mm ² /s hitherto: St (<i>Stokes</i>) 1 St = 1 cm ² /s cSt (<i>Centistokes</i>) 1 cSt = 1mm ² /s
Temperature, Heat			
Temperature Temp. interval	K <i>Kelvin</i>		<u>Basic unit</u> W.E.: °C (0°C ≈ 273.15 K; + 100 °C ≈ 373.15 K) with °C no decimal prefixes °K and grd not permitted any more after 1975
Specific heat capacity	J/kg K*		W.E.: kJ/kgK hitherto: kcal/kg °C Alteration of numerical values
Calorific value	J/kg* J/m ³ *		W.E.: kJ/kg, kJ/m ³ hitherto: kcal/kg, kcal/m ³ Alteration of numerical values

Remarks: * To be avoided with MTU-F

[] means: Altered, new symbols according to DIN 1304

Italics imply: Additional explanation, prefer symbols underlined

General



Dismantling



Group Dismantling and Assembly



Main Assembly



Test-Bed Run



Appendix



PREFACE

This **publication** is intended to inform about the MTU Diesel engine and aid in its operation, servicing and maintenance.

The section **Technical Data** contains details of the engine output, weight, settings and operational values etc.

The section **Description and Directions for Servicing** contains, besides the description of the engine components, instructions for performing inspections, servicing and maintenance.

As regards the inspection and maintenance operations which must be carried out at regular intervals, the time sequences of the maintenance schedule are valid.

Directions for operations of greater extent, which go beyond normal inspection and maintenance, are contained in the **Assembly Instructions**.

In the section **Operating Instruction** hints are given for operating the engine and its supervision before, during and after operation.

The section **Maintenance** lists the maintenance operations and a maintenance schedule.

The time schedule indicates at which intervals the maintenance operations have to be carried out.

The section **Hints for Trouble Shooting** names possible fault symptoms and gives hints on the cause and also the correction of the fault.

The section **Appendix** contains illustrations and drawings.

The model designation and specifications given by the manufacturers of the ancillaries and equipment mounted to the engine conform to the engine equipment at the time of issue of this publication.

These particulars may in no case be used for ordering spare parts.

For the ordering of spare parts is authorized exclusively the spare parts catalogue.

CONTENTS

- 1.1. MAIN DATA
- 1.2. MAJOR DIMENSIONS
- 1.3. ENGINE WEIGHT, OIL AND COOLING WATER
CAPACITY
- 1.4. OPERATIONAL DATA

